Closing the Romanized-Hindi Gap in Affect and Stress-Language Classification: A Controlled Multi-Seed Study of Pre-training Data and Multi-Task Sharing

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***Abstract*—Screening classifiers for sentiment, emotion, and stress-related language often rest on one shared multilingual encoder, yet Romanized Hindi mixed with English (Hinglish) remains harder than English. We ask what explains that gap and which changes reduce it while keeping the evaluation honest. Using three public sources (Dreaddit for English stress language, English and Romanized-Hindi sentiment data, and BRIGHTER for English and Hindi emotion), we freeze the splits with hashed membership lists, remove duplicates across splits, and compare a TF-IDF baseline, a BiLSTM, and MuRIL over three seeds with paired bootstrap intervals. Replacing MuRIL by the Hinglish-pre-trained HingRoBERTa raises Hinglish sentiment macro-F1 from 0.556 to 0.611 (+0.056, 95% CI [+0.032, +0.079]) with no significant change on English, and an exact-match scan found none of the test texts of five or more words in its pre-training corpus. A dedicated stress model raises PR-AUC from 0.852 to 0.875, whereas sharing one encoder across tasks helped none of them. Longer inputs gave no gain, and abstaining on the least confident 17% of stress posts raises macro-F1 from 0.783 to 0.828. Integrated Gradients attributions are faithful for English stress but only weakly so for Hinglish. The stress model is on par with the original Dreaddit logistic regression and slightly below its BERT-base result. On emotion, IndicBERT v2, the encoder of a SemEval-2025 Hindi system, beats MuRIL on our splits by +0.063 macro-F1 overall and +0.108 on Hindi, so our MuRIL emotion models are not the best available choice. A 50-case behavioral probe shows that all three model families read sarcasm at face value and assign emotions and stress to neutral sentences. The system is a research tool for language signals and does not diagnose any condition.**

***Index Terms*—code-mixed text, Hinglish, sentiment analysis, stress detection, multi-task learning, calibration, selective prediction, reproducibility.**

I. Introduction

Text classifiers for sentiment, emotion, and stress-related language are used to study how people write about their lives online. Most of them rest on a pre-trained multilingual encoder such as XLM-R [1] or MuRIL [2]. For many Indian users, much of this text is Romanized Hindi mixed with English, usually called Hinglish. Code-mixed benchmarks such as GLUECoS [3] and the SemEval-2020 sentiment task [4] exist to measure how well models handle it, and the HingBERT study [5] reports that models pre-trained on Roman-script Hinglish tweets do better on code-mixed tasks.

Published comparisons are hard to interpret for three reasons. Fine-tuned transformers vary across random seeds by amounts comparable to many reported gains [6], [7], yet single-seed results are common. The English to Hinglish gap is usually estimated across datasets that differ in platform, annotators, and class balance, so language is confounded with everything else. Finally, a pre-trained checkpoint may have seen the test text, and shared multi-task models are rarely compared with single-task ones under the same budget. For stress-related language the cost of a wrong conclusion is higher: models of mental health built on social media data generalize poorly across datasets [8], and the stress label used here records whether crowd workers judged a text to express stress, not a clinical state [9].

This paper studies one question: what explains the performance gap between English and Romanized-Hindi text classification, and which interventions reduce it while keeping the evaluation honest? We split it into four sub-questions. *RQ1:* does Hinglish-specific pre-training close part of the gap on held-out data? *RQ2:* does sharing one encoder across sentiment, emotion, and stress help or hurt each task? *RQ3:* do longer inputs help stress detection on long Reddit posts? *RQ4:* are the probabilities and token explanations of the resulting models trustworthy?

Our contributions are the following.

1) A reproducible protocol with frozen, hashed splits, duplicate removal across splits, validation-only model selection, three seeds, paired bootstrap intervals, and a pre-registered list of runs that may touch the test set.

2) Evidence that replacing MuRIL by HingRoBERTa raises Hinglish sentiment macro-F1 by +0.056 on test without a significant English loss, together with a contamination scan of its pre-training corpus.

3) A controlled single-task versus multi-task ablation over three unrelated datasets: no task gains from sharing, and stress PR-AUC falls.

4) Negative and mixed findings: longer stress inputs do not help, and Integrated Gradients explanations are faithful for English stress but weak for Hinglish.

5) Data findings in public benchmarks: the Hindi tweet configuration is Romanized and partly English, one of its validation splits scores near chance for every model, and the English emotion data has no disgust labels.

6) A 50-case behavioral probe of the three model families on sarcasm, contradictory emotions, Hinglish, and neutral text, released as a script.

7) A comparison on our frozen splits with IndicBERT v2, the encoder behind a 2025 shared-task system, which shows that MuRIL is the weaker encoder for emotion in both languages.

The system is a research screening classifier for language signals. It does not diagnose depression, anxiety, or any other condition, and its stress output is not a clinical probability.

II. Related Work

A. Stress and mental-health text

Dreaddit [9] provides Reddit post segments from stress-related communities, labeled for stress by crowd workers. Its authors report an F1 of 0.7980 for their best logistic regression and 0.8065 for fine-tuned BERT-base on the official test set (their Table 6). Chancellor and De Choudhury [10] review predictive techniques for mental health status on social media, Harrigian *et al.* [8] test whether such models generalize across datasets, and Benton *et al.* [11] give ethical protocols for this research. Those works motivate our restriction to language signals, our English-only validation claim, and our decision to keep all text out of the repository.

B. Multilingual and code-mixed models

XLM-R [1] covers 100 languages. MuRIL [2] is trained only on Indian-language corpora, plus pairs of documents and their translations or transliterations. HingBERT, HingMBERT, and HingRoBERTa [5] are pre-trained on L3Cube-HingCorpus, 52.93 million Roman-script Hinglish sentences scraped from Twitter. On the Roman-script GLUECoS sentiment task they report F1 scores of 66.43 for HingRoBERTa, 61.57 for XLM-R, and 58.42 for multilingual BERT (their Table 4). Barbieri *et al.* [12] released a multilingual tweet sentiment benchmark, UMSAB, whose Hindi portion we use, and pre-trained XLM-Twitter on tweets. Dakshina [13] and Aksharantar [14] document romanization variability and provide transliteration resources, which we leave to future work. IndicBERT v2 [15] is pre-trained on IndicCorp v2 for 23 Indic languages and English. At SemEval-2025 Task 11 [16], the indiDataMiner system [17] fine-tuned it on BRIGHTER and reported 87.37 F1 for Hindi emotion. We use it as a recent point of comparison in Section V-J.

C. Multi-task learning

MT-DNN [18] reports gains from multi-task fine-tuning on language-understanding tasks. Pruksachatkun *et al.* [19] study 110 intermediate-to-target task pairs and find that intermediate tasks needing high-level inference transfer best. Gradient conflicts between tasks are one proposed mechanism for negative transfer [20].

D. Training variance, calibration, and explanations

Dodge *et al.* [6] and Mosbach *et al.* [7] show that seeds and optimization choices change fine-tuning results substantially. Sun *et al.* [21] find that keeping the first 128 and last 382 tokens of long documents works best on two long-text datasets. Guo *et al.* [22] recommend temperature scaling, Desai and Durrett [23] study the calibration of pre-trained transformers, and selective classification [24] and maximum-probability baselines [25] let a model abstain. Integrated Gradients [26] is a common attribution method, ERASER [27] defines comprehensiveness and sufficiency measures, and Jacovi and Goldberg [28] argue that faithfulness should be evaluated explicitly. For significance testing in NLP we follow [29] and [30].

We did not find a study that compares Hinglish-specific pre-training, single-task and multi-task training, input length, calibration, and explanation faithfulness under one protocol. This paper does so on small public datasets.

III. Data

A. Sources

Table I lists the data. Stress comes from Dreaddit [9], which has an official train and test split. Sentiment combines the English and "hindi" portions of UMSAB [12], [31] with a community-built corpus of 3,190 manually labeled Hinglish YouTube comments from 23 videos [32], whose annotator count and agreement are not documented. Emotion uses the English and Hindi (Devanagari) portions of BRIGHTER [33], [16], annotated for six binary emotions. The three tasks come from different populations, so we never merge labels, and all comparisons across tasks are descriptive.

Table I. Data after Preprocessing (Train / Validation / Test)

Source	Language	Task	Train / Val / Test
Dreaddit	English	stress (2 classes)	2,391 / 425 / 715
UMSAB English	English	sentiment (3)	1,833 / 322 / 870
UMSAB "hindi"	Romanized	sentiment (3)	1,811 / 322 / 866
Hinglish YouTube	Romanized	sentiment (3)	2,210 / 477 / 477
BRIGHTER English	English	emotion (5 labels)	2,379 / 475 / 2,759
BRIGHTER Hindi	Devanagari	emotion (6 labels)	2,243 / 405 / 1,010

B. Preprocessing and leakage control

Text is normalized (NFKC, control characters removed) and URLs, e-mail addresses, phone numbers, and @mentions are replaced by placeholders before any model sees them. Exact duplicates are removed within each split. Rows in a lower-priority split that are exact or near duplicates (cosine similarity of at least 0.9 on punctuation-normalized character 4 to 5-gram TF-IDF vectors) of a higher-priority split are removed, with priority test, then validation, then train; test rows are otherwise untouched. Splits are grouped where metadata exists: Dreaddit validation is 15% of the training posts grouped by post identifier (train and test share no post), and the YouTube comments are split 70/15/15 by video, which gives only 23 groups. Because the official BRIGHTER development sets are small, emotion validation is the development set plus a seeded 12% of train. The final split membership is stored as text identifiers, split labels, and SHA-1 hashes of the cleaned text, with a SHA-256 manifest. Loading data fails if a processed table differs from the frozen list.

C. Data findings

Inspecting the files changed four assumptions. First, the UMSAB "hindi" configuration is written in Latin script: all

validation and test tweets, and all but one training tweet, have at most 5% Devanagari letters. Our language identifier labels 223 of its 866 test tweets as English. We therefore tag it Romanized Hindi and claim no Devanagari sentiment support. Second, the English BRIGHTER files have no disgust annotation at all, so disgust is treated as a missing label and masked in the loss and in every metric. Third, every BRIGHTER development and test text appears twice (as Track A and Track C copies with identical labels), which we collapsed, and ten corrupted rows were dropped. Fourth, on the UMSAB "hindi" validation split every model, including TF-IDF, scores near chance (about 0.35 macro-F1) although the same source reaches 0.55 to 0.62 on its test split. We could not determine the cause (label misalignment or a harder split) and therefore do not use that subset for model selection.

IV. Method

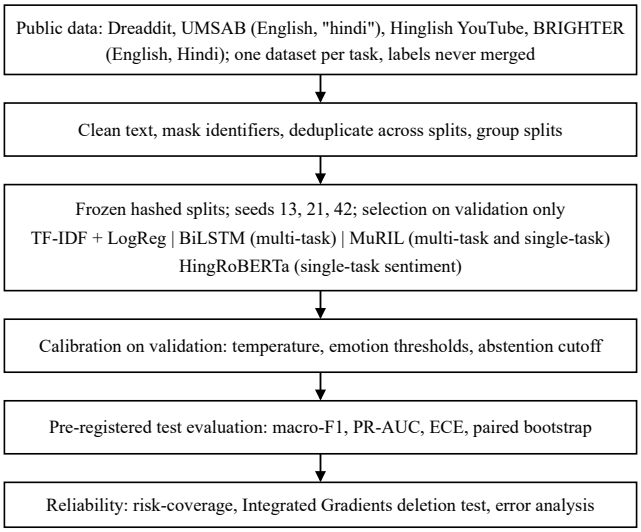


Fig. 1. Experimental pipeline. Test data is used only by the runs listed in the pre-registered plan.

A. Models

TF-IDF baseline. Word 1 to 2-gram and character 2 to 5-gram TF-IDF features (sublinear term frequency, at most 150,000 character features) feed one logistic regression per task. Emotion uses one binary regression per label and ignores rows where that label is missing. The regularization constant is chosen on validation from {0.5, 2, 8}. The model has 1.16 million parameters.

BiLSTM. A word-level tokenizer (vocabulary of at most 30,000 types, minimum frequency 2) feeds 200-dimensional embeddings trained from scratch, one bidirectional LSTM layer [34], [35] with 192 units per direction, and masked mean and max pooling. It has 3.18 million parameters and no pre-trained weights.

Transformers. MuRIL (237.95 million parameters with heads) and HingRoBERTa (278.4 million) are fine-tuned with a shared representation. For a sequence of encoder outputs $e_1, \ldots, e_n$ we pool $h = [\text{mean}(e_t); \max(e_t)]$ over non-padding tokens, and compute $z = \text{Dropout}(\text{GELU}(W_s h + b_s))$. A linear head per task maps z to logits: three classes for sentiment, six independent sigmoids for emotion, and two classes for stress. The multi-task model has all three heads and the single-task models have one, with identical initialization and hyperparameters otherwise.

The multi-task loss is

$$L = \lambda_{sent} L_{sent} + \lambda_{emo} L_{emo} + \lambda_{stress} L_{stress} \quad (1)$$

with all λ equal to 1. Sentiment and stress use cross-entropy. Emotion uses a masked binary cross-entropy

$$L_{emo} = \Sigma_{i,j} m_{ij} \ell_{ij} / \Sigma_{i,j} m_{ij} \quad (2)$$

where m_{ij} is 1 if label j is annotated for example i and 0 otherwise, and positive terms carry the weight $\min(5, \max(1, \sqrt{(neg/pos)}))$. Class weights are used for emotion only: the training imbalance ratios for sentiment (1.41) and stress (1.10) are below the threshold of 1.5 that we set in advance. Batches contain one task only, and batches from all tasks are shuffled together each epoch.

B. Training

Table II lists the settings. We use AdamW [36], PyTorch [37], the Transformers library [38], and scikit-learn [39]. The best-validation weights are restored for evaluation. The same hyperparameters are used for all transformer runs, and the only tuning is the logistic-regression constant.

Table II. Training Configuration

Setting	BiLSTM	MuRIL / HingRoBERTa
Optimizer	AdamW, 2e-3	AdamW, encoder 3e-5, heads 5e-4
Weight decay / clipping	0.01 / 1.0	0.01 / 1.0
Schedule	6% warm-up, linear decay	6% warm-up, linear decay
Batch size / epochs	32 / 12	16 / 3
Early stopping (patience)	3	2
Max length (sent./emo./stress)	64 / 64 / 128 tokens	64 / 64 / 128 tokens
Seeds / precision / device	13 / fp32 / CPU	13, 21, 42 / fp32 / CPU
Parameters	3.18 M	237.95 M / 278.44 M

The BiLSTM and TF-IDF baselines were trained with one seed (13); transformer comparisons use three seeds.

C. Calibration and selective prediction

For each task we fit one temperature T on the validation negative log-likelihood so that probabilities are softmax(z/T) [22]; emotion uses a sigmoid temperature and per-label thresholds searched on a grid from 0.1 to 0.9 that maximize validation F1. Expected calibration error (ECE) uses 15 equal-width bins on the top-label confidence, and we also report the Brier score. For selective prediction, confidence is the maximum calibrated probability, the abstention cutoff is the 20th percentile of validation confidence (a target coverage of 80%), and the cutoff is then applied unchanged to the test set. We also report the area under the error-coverage curve (AURC) [24].

D. Evaluation protocol

Sentiment and emotion are scored by macro-F1 (emotion over annotated label cells only), and stress by F1 of the positive class, PR-AUC, ROC-AUC, and ECE. Sentiment is reported separately for English tweets, Romanized-Hindi tweets, and

Hinglish YouTube comments. All selection uses validation data. Only the runs named in the plan touch the test set: the multi-task MuRIL model, the three single-task MuRIL models, and HingRoBERTa, which was evaluated once after it won on validation. The 256-token stress variants were compared on validation only. System pairs are compared by a paired bootstrap [29], [30]: we resample test examples 1,000 times, average each metric over the three seeds on the same resample for both systems, and report the 2.5th and 97.5th percentiles of the difference. This interval covers test-sample noise only, so seed standard deviations are reported separately.

E. Faithfulness test

The application serves Integrated Gradients [26] attributions over the input embeddings (16 steps, zero baseline). We test them with the comprehensiveness measure of ERASER [27]: for 60 validation examples per group, we delete the $k = \lceil 0.2n \rceil$ content tokens with the highest attribution, a random set of k tokens (mean of five draws), and the k lowest-attribution tokens, and measure the drop in the probability of the predicted class. Faithful attributions should give a larger drop for the top set than for the random set [28].

F. Reproducibility engineering

Each run stores its resolved configuration, package versions, git commit, per-epoch losses, calibrated per-example probabilities (without text), and timing. Training writes an atomic snapshot of weights, optimizer, scheduler, random state, and position within the epoch every ten minutes, so a machine restart costs at most that long and repeats no batch. A unit test covers interruption and resumption. Seeding happens before the model is built: an early version initialized the task heads before seeding, so "seed 13" differed between processes, and the regression is now tested. Our multi-task seed-13 run reproduces the earlier October run exactly at every epoch.

V. Experiments and Results

A. Setup and deviations

Apart from the comparison in Section V-J, no GPU was used, so every run used four CPU cores in single precision, roughly one day of compute in total for 21 runs. We therefore used three seeds instead of five and trained single-task stress models, which cost about a quarter of a multi-task run. A learning-rate grid, an XLM-R control, and normalization and transliteration experiments were not run, and the manual error taxonomy has not been annotated.

B. Baselines and the multi-task transformer

Table III compares the three model families from the first single-seed run. The BiLSTM does not beat TF-IDF: its paired differences from the baseline on test are -0.013 macro-F1 for sentiment (95% CI [-0.037, +0.010]) and -0.019 F1 for stress (CI [-0.048, +0.010]), and its validation F1 plateaus after about six epochs while training loss keeps falling. MuRIL gains +0.123 on sentiment (CI [+0.097, +0.147]) and +0.047 on stress F1 (CI [+0.018, +0.079]). Its stress PR-AUC is not higher than that of TF-IDF (-0.010, CI [-0.045, +0.026]), so its F1 advantage comes from the operating point and not from better ranking. It also needs 89 ms per request on CPU against 4 ms.

Table III. First Single-Seed Comparison (Test, Seed 13)

Model	Params	Sent. F1	Emo. F1	Stress F1	PR-AUC	ms
TF-IDF + LogReg	1.16 M	0.489	0.612	0.760	0.843	3.9
BiLSTM	3.18 M	0.475	0.561	0.742	0.821	3.9
MuRIL (multi-task)	237.95 M	0.611	0.690	0.807	0.830	89.2

F1 values are macro-F1 for sentiment and emotion and positive-class F1 for stress; ms is median CPU latency for all three heads on one short text.

C. Hinglish sentiment (RQ1)

Table IV and Fig. 2 compare single-task MuRIL with single-task HingRoBERTa, with the same splits, hyperparameters, and seeds. HingRoBERTa selected itself on validation, where its Hinglish gain was +0.078 [+0.052, +0.107]. On test it improves Hinglish macro-F1 from 0.556 to 0.611, a difference of +0.056 [+0.032, +0.079], and the gain appears on both Hinglish sources. English changes by +0.020 [-0.002, +0.044], so we cannot claim an English effect in either direction. The English to Hinglish gap falls from 0.128 to 0.093, a reduction of 27%.

Table IV. Sentiment Macro-F1 on Test, MuRIL vs. HingRoBERTa (3 Seeds)

Test subset (n)	MuRIL	HingRoBERTa	Difference [95% CI]
English tweets (870)	0.684 ± 0.004	0.704 ± 0.009	+0.020 [-0.002, +0.044]
Romanized-Hindi tweets (866)	0.561 ± 0.005	0.616 ± 0.034	+0.054 [+0.028, +0.083] *
Hinglish YouTube (477)	0.531 ± 0.006	0.602 ± 0.014	+0.071 [+0.026, +0.114] *
All Hinglish (1,343)	0.556 ± 0.005	0.611 ± 0.022	+0.056 [+0.032, +0.079] *
All sentiment (2,213)	0.606 ± 0.003	0.647 ± 0.013	+0.042 [+0.025, +0.058] *

± is the standard deviation over seeds. An asterisk marks a bootstrap interval that excludes zero. HingRoBERTa has 17% more parameters than MuRIL and a different tokenizer.

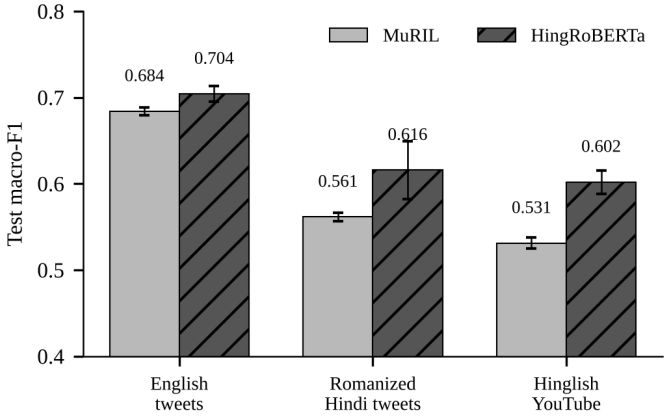


Fig. 2. Test macro-F1 by subset for single-task MuRIL and HingRoBERTa. Bars show means over three seeds and error bars show one standard deviation.

Most of the gain comes from fewer texts wrongly labeled negative. Pooling the three seeds' probabilities, MuRIL predicts negative for 131 positive and 127 neutral Hinglish test comments, and HingRoBERTa does so for 97 and 80. Negative recall is about unchanged. Inside the Romanized-Hindi tweets, which share a platform and annotation pipeline, tweets that our identifier tags as English score 0.642 (MuRIL)

and 0.676 (HingRoBERTa), and those tagged Romanized Hindi score 0.534 and 0.596. A gap of about 0.1 therefore remains within one source. The identifier is itself a classifier (recall 0.92 on Hinglish YouTube text and 0.99 on English), so this split is indicative.

Contamination. HingRoBERTa was pre-trained on tweets, so we scanned all 52,933,356 lines of a public re-upload of the corpus for exact matches (after normalization) with Hinglish validation and test texts of at least five words. None of the 1,149 test texts matched, and one of 675 validation texts did. Near duplicates were not checked, and the re-upload is not verified identical to the original release.

D. Stress detection and multi-task sharing (RQ2)

Table V gives the three-seed test results and Table VI the controlled comparison. A dedicated stress model has higher PR-AUC than the multi-task model (0.875 against 0.852). The multi-task minus single-task difference is -0.023 [-0.038, -0.008], and the F1 difference is -0.003 [-0.017, +0.012], so only ranking quality changes. Calibration moves the other way: ECE is 0.062 for the single-task model and 0.035 for the multi-task one. For sentiment and emotion, sharing neither helps nor hurts within our resolution. The three datasets differ in population, platform, and annotators, and some tasks transfer much better than others [19], so a positive effect was not guaranteed; the gains reported for related tasks [18] may not extend to this setting. We did not test gradient-conflict remedies [20], which the stress result would justify.

TABLE V. TEST RESULTS OVER THREE SEEDS (MEAN ± SD)

Model	Sent. F1	Emo. F1	Stress F1	Stress PR-AUC	Stress ECE
MuRIL multi-task	0.603 ± 0.008	0.684 ± 0.011	0.796 ± 0.014	0.852 ± 0.029	0.035 ± 0.002
MuRIL single-task	0.606 ± 0.003	0.682 ± 0.002	0.798 ± 0.005	0.875 ± 0.010	0.062 ± 0.023
HingRoBERTa single-task	0.647 ± 0.013	n/a	n/a	n/a	n/a

TABLE VI. MULTI-TASK MINUS SINGLE-TASK MURIL (TEST)

Metric	Single	Multi	Difference [95% CI]
Sentiment macro-F1	0.606	0.603	-0.002 [-0.014, +0.011]
Hinglish	0.556	0.559	+0.003 [-0.013, +0.020]
English	0.684	0.673	-0.011 [-0.028, +0.009]
Emotion macro-F1	0.682	0.684	+0.003 [-0.005, +0.010]
English	0.639	0.645	+0.006 [-0.002, +0.012]
Hindi	0.770	0.768	-0.002 [-0.012, +0.009]
Stress F1	0.798	0.796	-0.003 [-0.017, +0.012]
Stress PR-AUC	0.875	0.852	-0.023 [-0.038, -0.008] *

An asterisk marks an interval that excludes zero.

E. Input length (RQ3)

With a 128-token limit, 23.6% of Dreaddit posts are truncated (median length 103 tokens, 95th percentile 179). Table VII compares the original setting with a 256-token limit, with and without head and tail truncation (first 64 and last 190 tokens, following [21]). Neither variant helps on validation. Head and tail truncation is worse in macro-F1 (-0.020 [-0.037, -0.003]), and plain 256-token input is indistinguishable from the original (-0.007 [-0.024, +0.010]). The result differs from [21], whose gains came from news and review articles of several hundred tokens; our posts are shorter and their signal appears to be early. Per the protocol, neither variant was evaluated on test.

TABLE VII. STRESS INPUT LENGTH (VALIDATION, 3 SEEDS)

Setting	Macro-F1	PR-AUC	ROC-AUC
128 tokens, head	0.792 ± 0.014	0.883 ± 0.007	0.868 ± 0.005
256 tokens, head	0.785 ± 0.003	0.883 ± 0.010	0.868 ± 0.003
256 tokens, head + tail	0.772 ± 0.019	0.876 ± 0.022	0.859 ± 0.022

F. Probabilities and abstention (RQ4)

After temperature scaling, test ECE is 0.062 for single-task stress, 0.035 for the multi-task stress head, 0.053 for MuRIL sentiment, and 0.063 for HingRoBERTa sentiment. HingRoBERTa has the lower Brier score (0.485 against 0.521) and the higher ECE, so calibration and discrimination should be reported separately. Abstention helps every model (Fig. 3). With the cutoff fixed on validation, the stress model answers 83% of test posts at macro-F1 0.828 (against 0.783 on all posts), MuRIL sentiment answers 77% at 0.649 (against 0.606), and HingRoBERTa answers 74% at 0.702 (against 0.647). Coverage on test differs from the 80% target because the confidence distribution shifts between splits. A low-confidence output means the model is uncertain about the text; it says nothing about the author.

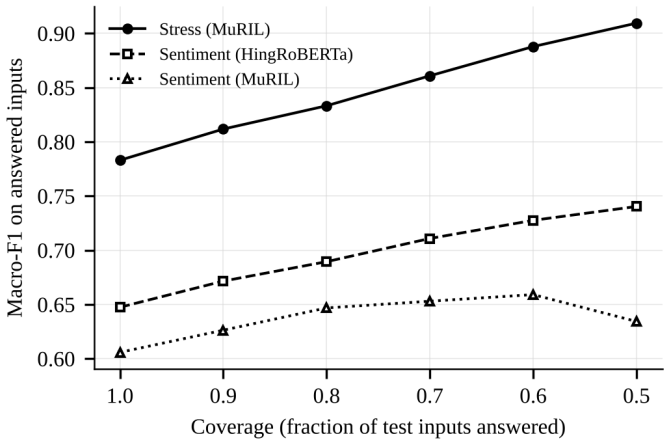


Fig. 3. Macro-F1 on answered test inputs as coverage falls, averaged over three seeds. The AURC is 0.104 for stress, 0.232 for HingRoBERTa sentiment, and 0.280 for MuRIL sentiment (lower is better).

G. Explanation faithfulness

Fig. 4 shows the deletion test. For English stress, deleting the top-attributed tokens lowers the predicted-class probability by 0.360 against 0.056 for random tokens (difference +0.305, bootstrap CI [+0.222, +0.386], top set larger in 92% of examples). English sentiment gives +0.155 (CI [+0.090, +0.224], 68% of examples). Hinglish sentiment gives only +0.074 (CI [+0.023, +0.125], 58% of examples), and for 42% of Hinglish examples the highlighted tokens matter no more than random ones. We measured comprehensiveness only, on 60 examples per group and one model, so these figures are only an initial check. The application now states this limit next to every explanation.

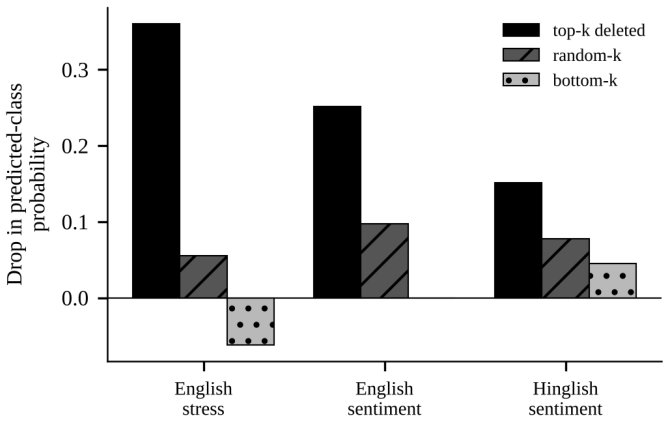


Fig. 4. Drop in predicted-class probability after deleting the top-k, random-k, or bottom-k tokens (k is 20% of content tokens; 60 validation examples per group).

H. Error analysis

On the Hinglish test set the error rate is 0.436 for MuRIL and 0.370 for HingRoBERTa, against 0.309 and 0.282 on English. Texts under eight words are harder (0.433 and 0.371) than texts of 8 to 39 words (0.372 and 0.323). Nearly all Dreaddit test posts are long (693 of 715 have 40 or more words), so the short-text behavior of the stress model is untested. The tweet language-identification split above and the near-chance validation subset (Section III-C) are the two data effects we can quantify. Categories such as spelling variation, code switching, and sarcasm need human annotation; we exported a sheet of 200 validation errors for that purpose but have not annotated it, so we report no category counts. Section V-I probes these phenomena with hand-written inputs instead.

I. Behavioral probe on hand-written edge cases

Test-set scores say little about failure modes that the datasets rarely contain. We therefore wrote 50 short inputs in nine categories (sarcasm, hidden sadness without emotion words, contradictory emotions, Hinglish code-switching, negative words with a positive emotion, neutral statements, ambiguity, short slang, and adversarial phrasing; 15 of them in Romanized Hindi) and sent each once to the application's prediction endpoint for the three model families. The emotion head has only six labels, so each expected label is mapped by a fixed rule to the nearest one: loneliness, numbness, and emptiness map to sadness, and numbness and emptiness also pass when no emotion fires. A contradictory case passes only if both emotions are detected, and a neutral case only if no emotion fires and the sentiment is neutral. 8 cases with no defensible single answer are not scored. The checkpoints were retrained for this probe with the released code and seed 13 on a GPU, and the MuRIL run stopped after one of its 3 epochs (validation mean macro-F1 0.601), so the probe describes these checkpoints and not the runs in Table III to Table VI.

TABLE VIII. BEHAVIORAL PROBE: PASSED OF SCORED CASES

Category	Scored	TF-IDF	BiLSTM	MuRIL
Sarcasm and fake positivity	5	0	0	1
Hidden sadness, no emotion words	5	3	2	4
Contradictory emotions	5	2	1	2
Hinglish code-switching	5	0	2	5
Negative words, positive emotion	5	3	2	2
Neutral statements	3	0	0	0
Ambiguity and context	2	0	1	0
Short text and slang	4	3	3	2
Adversarial phrasing	8	6	6	7
All scored cases	**42**	**17**	**17**	**23**
Two or more emotions (of 50)		21	37	34
Flagged as stress (of 50)		8	24	30

Scoring is a fixed rule, not a human judgment. Per-case inputs, outputs, and checks are in reports/results/edge_case_probe.json.

MuRIL passes 23 of 42 scored cases, against 17 for TF-IDF and 17 for the BiLSTM. It handles all five Hinglish cases and seven of eight adversarial ones, where the other models pass at most two and six. All three read sarcasm at face value: the English sarcastic inputs mostly receive joy and positive sentiment, and only MuRIL catches one case, the Hinglish one. No model passes a neutral case. "I am sitting alone in my room" receives fear and sadness and is flagged as stress by every model, which fits the untested short-text behavior of the stress head noted above. Positive and negative emotions are rarely detected together: of the four cases that need joy plus a negative emotion, only one TF-IDF output has both. Emotions also over-fire: two or more labels appear on 21, 37, and 34 of 50 inputs, and fear on 27 to 30. With 50 cases scored by a rule, the probe gives qualitative evidence only, with no accuracy estimate.

J. Comparison with a 2025 system's encoder

Most of our comparison points date from 2019 to 2022. To add a recent one, we ran the encoder of the indiDataMiner system [17] from SemEval-2025 Task 11, IndicBERT v2 [15] (ai4bharat/IndicBERTv2-MLM-only, 278 million parameters), on our own emotion splits. It uses the same single-task head, loss, hyperparameters, three seeds, and test protocol as single-task MuRIL, and was added to the pre-registered list before it was run. Unlike the MuRIL runs it was trained on a GPU (an RTX 3050, about 35 minutes per seed), still in single precision. Table IX gives the result.

TABLE IX. EMOTION MACRO-F1 ON TEST, MURIL VS. INDICBERT V2 (3 SEEDS)

Scope	MuRIL	IndicBERT v2	Difference [95% CI]
All	0.682 ± 0.002	0.744 ± 0.003	+0.063 [+0.051, +0.074]
English	0.639 ± 0.004	0.676 ± 0.004	+0.037 [+0.027, +0.047]
Hindi	0.770 ± 0.008	0.878 ± 0.002	+0.108 [+0.091, +0.126]

Paired bootstrap over test examples (1,000 resamples) on seed-averaged scores. Differences are IndicBERT v2 minus MuRIL.

IndicBERT v2 is better in both languages and every interval excludes zero. The gain is largest on Hindi (+0.108), where it reaches 0.878, above the published RoBERTa baseline of 0.855 and close to the 87.37 F1 that [17] report, although their test protocol differs from ours. On English the gain is smaller (+0.037) and the score stays below the 0.708 baseline. The hardware differs between the two models, which

can shift results slightly, but not by margins of this size. IndicBERT v2 itself was released in 2022 and 2023; the 2025 link is the shared-task system that used it. The result means our MuRIL emotion numbers reflect the encoder choice, and an emotion model built on IndicBERT v2 should replace them.

VI. Comparison with Published Results

Our test results and the published numbers, read from the original tables, are in Table X. Only the Dreaddit comparison uses an identical test set and a standard protocol; the others differ in training data, baselines, or the removal of duplicate test rows, so they serve only as context.

Table X. Published Results and This Work

Task	Published	Ours (test)
Stress, Dreaddit [9]	BERT-base F1 0.8065; best LogReg 0.7980	single-task F1 0.798; multi-task 0.796
Hinglish sentiment gain [5]	HingRoBERTa 66.43 vs XLM-R 61.57 F1 (+4.86), GLUECoS	+0.056 over MuRIL, different data
UMSAB Hindi [12]	XLM-Twitter 56.39 ± 1.60; XLM-R 53.39 ± 2.00	HingRoBERTa 0.616; MuRIL 0.561
UMSAB English [12]	XLM-Twitter 70.63 ± 1.04; XLM-R 68.52 ± 1.42	HingRoBERTa 0.704; MuRIL 0.684
BRIGHTER English [16]	RoBERTa baseline 0.708; top system 0.823	0.639 single; 0.645 multi; IndicBERT v2 0.676
BRIGHTER Hindi [16]	RoBERTa baseline 0.855; top system 0.926	0.770 single; 0.768 multi; IndicBERT v2 0.878

UMSAB rows: the published numbers come from one multilingual model trained on all eight languages (their Table 7), whereas ours train on English tweets, Romanized-Hindi tweets, and Hinglish YouTube comments; our test sets are the same files (the Hindi file after removing 4 duplicate tweets). BRIGHTER: our test sets have duplicate Track A and C copies removed (2,759 and 1,010 texts), English lacks disgust labels, and published baselines were trained on the full training sets.

On Dreaddit our stress model is on par with the original paper: F1 0.798 against 0.7980 for their best logistic regression and 0.8065 for BERT-base, on the same official test posts. The size of the Hinglish gain (+5.6 macro-F1 points over MuRIL) is similar to the +4.9 F1 points that [5] report over XLM-R on a different benchmark, which supports the claim that Hinglish pre-training helps code-mixed sentiment. On the shared UMSAB Hindi test file HingRoBERTa scores about five points above the published XLM-Twitter figure, but the training data differs, so we treat it as indicative only. Our MuRIL emotion models are below the published baselines in both languages. We trained them for three epochs on CPU, with validation carved out of the training data, and did not tune them. With the same budget, IndicBERT v2 passes the Hindi baseline but not the English one (Section V-J).

VII. Discussion and Limitations

The measured effects are modest. Hinglish pre-training lowers the Hinglish error without a significant English cost, which suggests that part of the English to Hinglish gap in MuRIL comes from its pre-training data and not from the task. Because HingRoBERTa also has a different tokenizer and 17% more parameters, we cannot separate pre-training data from those factors, and an XLM-R control would help. The remaining gap of about 0.09 and the within-source gap of about 0.1 mean that data quality, label noise, and the English content of the "hindi" tweets still account for part of what looks like a language effect.

Stress detection did not benefit from any change we tried except isolating it from the other tasks, and even that changed ranking but not F1, and made calibration worse. The stress label is a crowd judgment of English Reddit text, so none of these results transfer to another language, platform, or population without new labeled data. We did not evaluate stress on Hindi or Hinglish and the application flags those outputs as unvalidated.

The served application still uses the multi-task MuRIL checkpoint. HingRoBERTa, the single-task stress model, and abstention were evaluated offline and are not yet served, so the application does not reflect the best configurations reported here. The behavioral probe adds that sarcasm, mixed emotions, and neutral statements remain failure modes for every model family. Fixing them needs labeled data for those phenomena.

The study has further limits. We ran three seeds on a CPU with a three-epoch budget, and for several runs the best validation epoch was the last, so the models may be under-trained; no learning-rate search was run. The bootstrap intervals cover test-sample noise only. The Hinglish data is small, the YouTube set has no documented annotator agreement, and one validation subset is unreliable. The contamination scan used exact matches on a re-upload of the corpus. Spelling normalization, transliteration [13], [14], and a human-annotated error taxonomy remain untested. Training times recorded for resumed runs count only the final process.

VIII. Conclusion

Under a controlled, multi-seed protocol on public data, a Hinglish-pre-trained encoder narrowed the English to Hinglish sentiment gap by about a quarter, sharing one encoder across unrelated tasks gave no benefit, and longer stress inputs did not help. Abstention and a faithfulness check show when the outputs can be trusted: abstaining raises macro-F1 on the answered inputs, and the token attributions should not be trusted for Hinglish text. A behavioral probe shows that sarcasm and neutral statements remain open failure modes. The next steps are serving HingRoBERTa and the single-task stress model, an emotion model built on IndicBERT v2, which beat MuRIL on both emotion test sets, a GPU run with five seeds and a learning-rate search, a human-annotated Hinglish error taxonomy, and a test of transliteration and spelling normalization. We release the configurations, frozen splits, per-run outputs, and analysis scripts so that these experiments can be repeated.

Ethics and Data Availability

This work is a research screening study of language signals. It makes no diagnosis, and its probabilities are statistical model outputs. The datasets are public Reddit, Twitter, and YouTube text of a sensitive kind, so the repository holds no raw text: it contains frozen split identifiers with text hashes, configurations, per-example probabilities, and code. The repository is public at

https://github.com/faizdevx/Multilingual-Mental-Health-Risk- and also holds earlier HTML and PDF versions of this paper by the same authors. Identifiers in the text are masked before modeling, and the application neither logs nor stores submitted text. Crisis information in the application links to two international helpline directories and contains no telephone numbers. Source licenses are summarized in the repository and should be checked before any redistribution.

Acknowledgment

The authors thank the creators of Dreaddit, UMSAB, BRIGHTER, the Hinglish YouTube corpus, MuRIL, and HingRoBERTa for releasing their data and models.

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